

CHAPTER 1

RELATIONS AND FUNCTIONS

VERY SHORT ANSWER TYPE QUESTIONS (1 MARK)

1. If A is the set of students of a school then write, which of following relations are. (Universal, Empty or neither of the two).

$$R_1 = \{(a, b) : a, b \text{ are ages of students and } |a - b| \geq 0\}$$

$$R_2 = \{(a, b) : a, b \text{ are weights of students, and } |a - b| < 0\}$$

$$R_3 = \{(a, b) : a, b \text{ are students studying in same class}\}$$

2. Is the relation R in the set $A = \{1, 2, 3, 4, 5\}$ defined as $R = \{(a, b) : b = a + 1\}$ reflexive?

3. If R , is a relation in set N given by

$$R = \{(a, b) : a = b - 3, b > 5\},$$

then does elements $(5, 7) \in R$?

4. If $f : \{1, 3\} \rightarrow \{1, 2, 5\}$ and $g : \{1, 2, 5\} \rightarrow \{1, 2, 3, 4\}$ be given by

$$f = \{(1, 2), (3, 5)\}, g = \{(1, 3), (2, 3), (5, 1)\}$$

Write down gof .

5. Let $g, f : R \rightarrow R$ be defined by

$$g(x) = \frac{x+2}{3}, f(x) = 3x - 2. \text{ Write } \text{fog}.$$

6. If $f : R \rightarrow R$ defined by

$$f(x) = \frac{2x-1}{5}$$

be an invertible function, write $f^{-1}(x)$.

7. If $f(x) = \frac{x}{x+1} \forall x \neq -1$, Write $fo f(x)$.

8. Let $*$ is a Binary operation defined on R , then if

$$(i) \quad a * b = a + b + ab, \text{ write } 3 * 2$$

(ii) $a * b = \frac{(a+b)^2}{3}$, Write $(2 * 3) * 4$.

9. If $n(A) = n(B) = 3$, Then how many bijective functions from A to B can be formed?
10. If $f(x) = x + 1$, $g(x) = x - 1$, Then $(g \circ f)(3) = ?$
11. Is $f : N \rightarrow N$ given by $f(x) = x^2$ is one-one? Give reason.
12. If $f : R \rightarrow A$, given by

$$f(x) = x^2 - 2x + 2$$
is onto function, find set A .
13. If $f : A \rightarrow B$ is bijective function such that $n(A) = 10$, then $n(B) = ?$
14. If $n(A) = 5$, then write the number of one-one functions from A to A .
15. $R = \{(a, b) : a, b \in N, a \neq b \text{ and } a \text{ divides } b\}$. Is R reflexive? Give reason?
16. Is $f : R \rightarrow R$, given by $f(x) = |x - 1|$ is one-one? Give reason?
17. $f : R \rightarrow B$ given by $f(x) = \sin x$ is onto function, then write set B .
18. If $f(x) = \log\left(\frac{1+x}{1-x}\right)$, show that $f\left(\frac{2x}{1+x^2}\right) = 2f(x)$.
19. If $*$ is a binary operation on set Q of rational numbers given by $a * b = \frac{ab}{5}$ then write the identity element in Q .
20. If $*$ is Binary operation on N defined by $a * b = a + ab \forall a, b \in N$. Write the identity element in N if it exists.

SHORT ANSWER TYPE QUESTIONS (4 Marks)

21. Check the following functions for one-one and onto.

(a) $f : R \rightarrow R, f(x) = \frac{2x-3}{7}$

(b) $f : R \rightarrow R, f(x) = |x + 1|$

(c) $f : R - \{2\} \rightarrow R, f(x) = \frac{3x-1}{x-2}$

- (d) $f : R \rightarrow [-1, 1], f(x) = \sin^2 x$
22. Consider the binary operation $*$ on the set $\{1, 2, 3, 4, 5\}$ defined by $a * b = \text{H.C.F. of } a \text{ and } b$. Write the operation table for the operation $*$.
23. Let $f : R - \left\{ \frac{-4}{3} \right\} \rightarrow R - \left\{ \frac{4}{3} \right\}$ be a function given by $f(x) = \frac{4x}{3x+4}$. Show that f is invertible with $f^{-1}(x) = \frac{4x}{4-3x}$.
24. Let R be the relation on set $A = \{x : x \in Z, 0 \leq x \leq 10\}$ given by $R = \{(a, b) : (a - b) \text{ is multiple of } 4\}$, is an equivalence relation. Also, write all elements related to 4.
25. Show that function $f : A \rightarrow B$ defined as $f(x) = \frac{3x+4}{5x-7}$ where $A = R - \left\{ \frac{7}{5} \right\}, B = R - \left\{ \frac{3}{5} \right\}$ is invertible and hence find f^{-1} .
26. Let $*$ be a binary operation on Q . Such that $a * b = a + b - ab$.
- Prove that $*$ is commutative and associative.
 - Find identify element of $*$ in Q (if it exists).
27. If $*$ is a binary operation defined on $R - \{0\}$ defined by $a * b = \frac{2a}{b^2}$, then check $*$ for commutativity and associativity.
28. If $A = N \times N$ and binary operation $*$ is defined on A as $(a, b) * (c, d) = (ac, bd)$.
- Check $*$ for commutativity and associativity.
 - Find the identity element for $*$ in A (If it exists).
29. Show that the relation R defined by $(a, b) R(c, d) \Leftrightarrow a + d = b + c$ on the set $N \times N$ is an equivalence relation.
30. Let $*$ be a binary operation on set Q defined by $a * b = \frac{ab}{4}$, show that
- 4 is the identity element of $*$ on Q .

(ii) Every non zero element of Q is invertible with

$$a^{-1} = \frac{16}{a}, \quad a \in Q - \{0\}.$$

31. Show that $f: R_+ \rightarrow R_+$ defined by $f(x) = \frac{1}{2x}$ is bijective where R_+ is the set of all non-zero positive real numbers.
32. Consider $f: R_+ \rightarrow [-5, \infty)$ given by $f(x) = 9x^2 + 6x - 5$ show that f is invertible with $f^{-1} = \frac{\sqrt{x+6}-1}{3}$.
33. If $*$ is a binary operation on R defined by $a * b = a + b + ab$. Prove that $*$ is commutative and associative. Find the identify element. Also show that every element of R is invertible except -1 .
34. If $f, g: R \rightarrow R$ defined by $f(x) = x^2 - x$ and $g(x) = x + 1$ find $(f \circ g)(x)$ and $(g \circ f)(x)$. Are they equal?
35. $f: [1, \infty) \rightarrow [2, \infty)$ is given by $f(x) = x + \frac{1}{x}$, find $f^{-1}(x)$.
36. $f: R \rightarrow R, g: R \rightarrow R$ given by $f(x) = [x], g(x) = |x|$ then find

$$(f \circ g)\left(\frac{-2}{3}\right) \text{ and } (g \circ f)\left(\frac{-2}{3}\right).$$

ANSWERS

1. R_1 : is universal relation.
 R_2 : is empty relation.
 R_3 : is neither universal nor empty.
2. No, R is not reflexive.
3. $(5, 7) \notin R$
4. $g \circ f = \{(1, 3), (3, 1)\}$
5. $(f \circ g)(x) = x \quad \forall \quad x \in R$

6. $f^{-1}(x) = \frac{5x+1}{2}$
7. $(f \circ f)(x) = \frac{x}{2x+1}, x \neq -\frac{1}{2}$
8. (i) $3 * 2 = 11$
(ii) $\frac{1369}{27}$
9. 6
10. 3
11. Yes, f is one-one $\because \forall x_1, x_2 \in N \Rightarrow x_1^2 = x_2^2$.
12. $A = [1, \infty)$ because $R_f = [1, \infty)$
13. $n(B) = 10$
14. 120.
15. No, R is not reflexive $\because (a, a) \notin R \forall a \in N$
16. f is not one-one functions
 $\because f(3) = f(-1) = 2$
 $3 \neq -1$ i.e. distinct element has same images.
17. $B = [-1, 1]$
19. $e = 5$
20. Identity element does not exist.
21. (a) Bijective
(b) Neither one-one nor onto.
(c) One-one, but not onto.
(d) Neither one-one nor onto.

22.

*	1	2	3	4	5
1	1	1	1	1	1
2	1	2	1	2	1
3	1	1	3	1	1
4	1	2	1	4	1
5	1	1	1	1	5

24. Elements related to 4 are 0, 4, 8.

25. $f^{-1}(x) = \frac{7x+4}{5x-3}$

26. 0 is the identity element.

27. Neither commutative nor associative.

28. (i) Commutative and associative.

(ii) (1, 1) is identity in $N \times N$

33. 0 is the identity element.

34. $(f \circ g)(x) = x^2 + x$

$(g \circ f)(x) = x^2 - x + 1$

Clearly, they are unequal.

35. $f^{-1}(x) = \frac{x + \sqrt{x^2 - 4}}{2}$

36. $(f \circ g)\left(\frac{-2}{3}\right) = 0$

$(g \circ f)\left(\frac{-2}{3}\right) = 1$